

Railway Performance & Delay Command Center

README / Project Documentation

Industry: Indian Railway Industry **Technology Stack:** MySQL + Power BI + DAX + ETL

Project Title

Railway Performance & Delay Command Center

Project Overview

Railway Performance & Delay Command Center is a data analytics and business intelligence project developed to monitor railway operations, passenger activity, revenue, train punctuality, delays, cancellations, stations, and routes.

The project converts operational data into an interactive Power BI command center so users can quickly identify performance problems, compare routes and stations, understand delay patterns, and support data-driven operational decisions.

The solution follows an end-to-end analytics workflow: Data Preparation → MySQL Database → ETL & Data Cleaning → Power BI Data Model → DAX Measures → Dashboard Development → Business Insights.

Project Objectives

- Monitor overall railway operational performance.
- Track train punctuality and average delays.
- Analyze passenger volumes and revenue.
- Identify high-delay trains, routes, and stations.
- Analyze cancellation trends and cancellation percentages.
- Understand delay reasons and their operational impact.
- Compare operations and punctuality across zones.
- Provide an interactive command center for data-driven operational decisions.
- Demonstrate an end-to-end analytics workflow using SQL, ETL, Power BI, and DAX.

Tools & Technologies

Tool / Technology	Purpose
MySQL	Relational database creation, table management, relationships, and SQL-based data storage
Power BI Desktop	Data modeling, interactive visualizations, dashboard development, and reporting
Power Query / ETL	Data extraction, transformation, cleaning, type correction, and loading
DAX	Calculated measures and business KPIs
Microsoft Excel / CSV	Initial dataset preparation and data exchange during ETL

Dataset Description

The project uses a synthetic railway operations dataset designed to represent realistic Indian railway operational scenarios.

The dataset includes train operations, train attributes, stations, routes, passengers, revenue, delay information, cancellation information, travel dates, operational attributes, and delay reason categories.

Dataset Source: The dataset was created as synthetic project data based on real-world railway operational patterns. It is intended for analytics, dashboard development, and academic/project demonstration. It is not an official Indian Railways operational dataset.

The data was intentionally prepared with practical data-quality issues so the project demonstrates a complete ETL and data-cleaning workflow.

SQL Analysis

A MySQL database named **railway_performance_db** was created to organize the railway data in a relational structure.

The database separates master information from operational records and uses primary-key and foreign-key relationships to maintain data integrity.

The Train_Operations table acts as the central operational table, while Trains, Stations, Routes, and Delay_Reasons provide descriptive/master information for analysis.

SQL/database work includes database and table creation, relational data organization, operational data storage, primary-key/foreign-key relationships, and preparation of structured data for Power BI.

Table	Purpose	Main Information
Trains	Train master data	Train ID, train name, train attributes
Stations	Station master data	Station ID, station name, station details
Routes	Route information	Route ID and route/station relationships
Delay_Reasons	Delay classification	Delay reason categories
Train_Operations	Core fact/transaction table	Travel date, train, route, origin, destination, passengers, revenue, delay, cancellation details

Power BI Dashboard

The Power BI report is designed as four focused dashboard pages, with each page serving a specific business purpose.

Page 1 – Executive Overview

KPI cards for Total Operations, Total Passengers, Total Revenue, Average Delay, ON Time %, and Cancellation %, with Monthly Average Delay, Monthly Revenue, Operations by Delay Reason, and Operations by Zone visuals.

Page 2 – Train Performance

Train-level analysis using Average Delay, Cancellation %, Total Revenue, Total Passengers, and ON Time %.

Page 3 – Station & Route Analysis

Route, station, and zone analysis using Total Operations, Average Delay, and ON Time %.

Page 4 – Delay & Cancellation Analysis

Average Delay by Hour and Day of Week, Delay Reason, Cancellation Status, monthly Cancellation %, and Total Cancellations.

Dashboard Visuals

- Total Cancellation by Month – clustered column chart
- Cancellation % Trend (Monthly) – line chart
- Cancellation Status Distribution – donut chart
- Delay Reason Distribution – donut chart
- Average Delay by Day of Week – clustered column chart
- Average Delay by Hour – line chart
- Top 10 Stations by Total Operations – clustered bar chart
- Top 10 Routes by Average Delay – clustered bar chart
- Operations by Zone – donut chart
- ON Time % by Zone – clustered bar chart
- Top 10 Stations by Average Delay – clustered bar chart
- Top 10 Routes by Total Operations – clustered bar chart
- Top 10 Trains by ON Time % – clustered bar chart
- Total Passengers by Train_Name – clustered column chart
- Total Revenue by Train_Name – clustered column chart
- Top 10 Trains by Cancellation % – clustered bar chart
- Top 10 Trains by Average Delay – clustered bar chart
- Operations by Delay Reason – clustered bar chart
- Monthly Revenue – line chart
- Monthly Average Delay – line chart

Key KPIs

KPI / Measure	Dashboard Usage
Total Operations	Executive Overview; Operations by Zone; Operations by Delay Reason; route/station analysis
Total Passengers	Executive Overview; Total Passengers by Train Name
Total Revenue	Executive Overview; Monthly Revenue; Total Revenue by Train Name
Average Delay	Executive Overview; monthly, train, route, station, hour, and day-of-week analysis
ON Time %	Executive Overview; ON Time % by Train Name and Zone
Cancellation %	Executive Overview; train-level analysis; monthly trend; Delay & Cancellation KPI
Total Cancellations	Delay & Cancellation analysis and monthly cancellation analysis

Key Insights

The dashboard is designed to help users identify and investigate overall railway operational performance.

It supports analysis of average delay patterns across trains, routes, stations, hours, and days; trains with lower ON Time % and higher cancellation %; routes and stations with higher average delays; zone-level operational volume and punctuality; monthly revenue and operational trends; major delay reason categories; monthly cancellation trends; and high-volume stations and routes.

Note: The exact numerical findings should be read directly from the Power BI dashboard because the documentation describes the dashboard structure and measures but does not provide final KPI values.

Project Workflow

End-to-end workflow

```
Raw / Prepared Railway Data
↓
Excel / CSV
↓
MySQL Database
↓
Relational Tables & Relationships
↓
Power Query / ETL
↓
Data Cleaning & Validation
↓
Power BI Data Model
↓
DAX Measures & KPIs
↓
Interactive Power BI Dashboard
↓
Business Analysis & Insights
```

ETL Steps

- Extract – Import railway operational and supporting master tables.
- Transform – Select required columns, correct data types, standardize text, and prepare analytical fields.
- Validate – Check nulls, duplicates, invalid values, relationships, and inconsistent categories.
- Load – Apply cleaned query results to the Power BI data model for visualization and DAX analysis.

Data Cleaning

- Handling duplicate records where applicable.
- Handling blank/null values in important fields.
- Correcting data types for dates, numbers, percentages, and categorical columns.
- Standardizing train, station, route, and reason names.
- Checking inconsistent delay and cancellation values.
- Validating primary-key and foreign-key relationships.
- Checking outliers and invalid operational values.

Repository Structure

```
Railway-Performance-Delay-Command-Center/
├── README.md
├── PowerBI/
│   ├── Railway_Performance_Delay_Command_Center.pbix
│   └── SQL/
│       ├── database_creation.sql
│       ├── table_creation.sql
│       └── analysis_queries.sql
├── Data/
│   ├── trains.csv
│   ├── stations.csv
│   └── routes.csv
```

- ■■■ delay_reasons.csv
- ■■■ train_operations.csv
- Documentation/
- ■■■ Project_Documentation.docx
- Screenshots/
- executive_overview.png
- train_performance.png
- station_route_analysis.png
- delay_cancellation_analysis.png

Update filenames above to match the actual files included in the GitHub repository.

Dashboard Screenshots

Add the Power BI dashboard screenshots to the Screenshots/ folder and reference them in the repository README.

How to Run the Project

1. Clone the repository.
2. Create the MySQL database railway_performance_db and create/populate the required tables using the SQL scripts.
3. Load the project CSV/Excel files into MySQL according to the database schema.
4. Open PowerBI/Railway_Performance_Delay_Command_Center.pbix in Power BI Desktop.
5. Configure the MySQL database connection under Home → Transform data / Data source settings.
6. Refresh the Power BI model after the connection is configured.
7. Navigate through the four report pages: Executive Overview, Train Performance, Station & Route Analysis, and Delay & Cancellation Analysis.
8. Use the dashboard slicers to investigate railway operations, delays, punctuality, revenue, passengers, routes, stations, and cancellations.

Dashboard filters/slicers: Zone, Destination Station, Origin Station, Train Type, Train Name, and Travel Date.

Author

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Project: Railway Performance & Delay Command Center

Domain: Data Analytics / Business Intelligence

Technology: MySQL | Power BI | DAX | Power Query / ETL